OPEN-SET PERSON IDENTIFICATION

OpenID Vision

Know when to match. Know when to reject.

The face decides when it can be trusted. The body supports it. A human is asked when the system is not sure.

MATCH UNKNOWN REVIEW

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The problem: look-alikes get in

Most person re-identification systems compare **clothing and build**. It is like a guard who only checks uniforms.

A **different person in a similar jacket** is accepted as an enrolled one. That wrong "yes" - a **false accept** - is the failure that matters most in access and security work.

PS-1 asks for exactly this setting: an **open set** (the person may never have been enrolled) with **low variance between people** (many look alike).

In our test, a clothing-only matcher accepted **51 of 60 strangers**.



Illustration: same jacket, different faces

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The idea: face first, body second, a human when unsure

MATCH

"This is enrolled person X."

The evidence passed an explicit rule. The door opens.

UNKNOWN

"Nobody enrolled is supported."

Never seen, or not similar enough. The door stays shut. Not an accusation - unknown people are never saved or tracked.

REVIEW

"Not sure - a person should look."

Blurry, no usable face, or a borderline score. The guard calls a supervisor.

A **nearest neighbour is never accepted just because it is nearest**. Clothing can support a face match, but can never turn a mismatching face into a MATCH.

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What happens to one photo

1. **Find the people**
   a detector boxes each person
2. **Check quality**
   blurry, tiny, cut off? then not sure
3. **Find and judge the face**
   big, sharp, facing the camera?
4. **Two fingerprints**
   face: 128 numbers · body: 512
5. **Compare with everyone enrolled**
   a face score and a body score each
6. **Decide with a written rule**
   MATCH · UNKNOWN · REVIEW
7. **Record it**
   scores, rule version, reason

Every number on the screen is computed by the server and saved with the rule version that produced it, so any decision can be explained and exported later.

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The decision rule

SITUATION	WHAT DECIDES	RESULT
Clear face, enrolled person	**the face** reaches the threshold	MATCH
Clear face, **stranger - even in the same clothes**	**the face** does not match	UNKNOWN
Face score just below the threshold	too close to call	REVIEW
No usable face (back view, far, blurred)	the **body** - weaker evidence, so a person is asked	REVIEW
Somebody enrolled without a face on file matches strongly by body	a face cannot rule them out	REVIEW

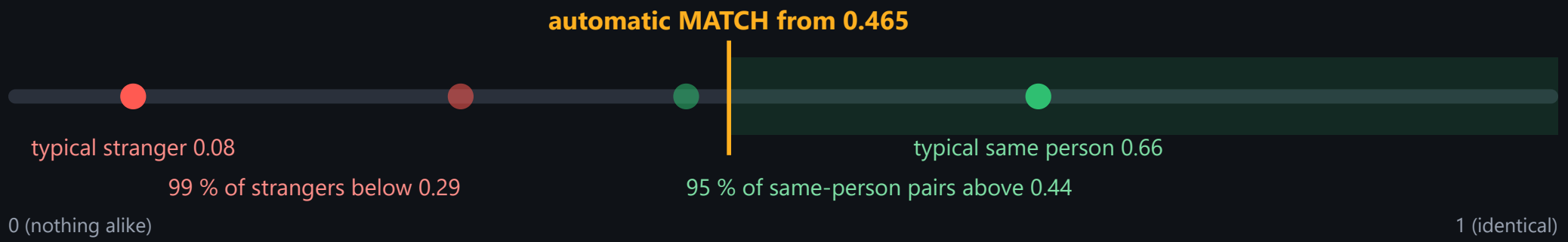
Face threshold **0.465** (automatic MATCH) · review band **0.10** below it · body threshold **0.70**. All are labelled *unvalidated* for any particular camera, and visible on every screen.

Proof 1 - the face stage on a standard public test

98.9 %
accuracy, 10-fold protocol
(± 0.35 %)

6,000
face pairs
(LFW benchmark)

7,701/7,701
pictures where a face
was found



Honest limit: LFW is easy (frontal, well-lit public figures). It says nothing about other cameras, masks, lighting or demographic groups.

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Proof 2 - why 0.465: a rule, not a guess

Rule fixed before looking at any result: **automatic MATCH only where strangers reach that score at most once in 100,000 comparisons**. Measured on **4,498,500 pairs of different people**.

FACE SCORE AT LEAST	STRANGERS REACHING IT (PER 100,000 PAIRS)	SAME-PERSON PAIRS ACCEPTED
0.363 model card default	81	97.7 %
0.4	19	96.8 %
0.45	1.8	94.5 %
0.465 OUR DEFAULT	1.0	93.4 %
0.5	0.4	90.5 %

With N enrolled people and about R photos each, the chance that one stranger is wrongly accepted is roughly N × R × 0.00001 - small, not zero, and it grows with the gallery. The 0.363 → 0.465 move removes about 99 % of the strangers that would get through.

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Proof 3 - the whole system, end to end

10 people enrolled through the normal interface, then 20 new photos of them and 60 photos of strangers. Same photos, two policies.

Strangers wrongly ACCEPTED (lower is better)



Enrolled people recognised correctly (higher is better)



Honest limit: these LFW head-and-shoulders photos are outside what the clothing model was designed for (full-body pedestrian crops), so the left bars are its worst case - this shows **what the face adds**, not a fair benchmark of the body model. The default threshold was fixed beforehand (previous slide), not tuned on these 60 strangers. Small sample.

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Proof 4 - the same suit, a different person

The home-page example: 13 public-domain photos with known right answers, including **3 strangers in the same orange flight suit** as an enrolled person.

Strangers in the same suit wrongly ACCEPTED (lower is better)

Clothes only	**3 of 3**
Face + body	**0 of 3**

Known people recognised (higher is better)

Clothes only	**0 of 5**
Face + body	**5 of 5**

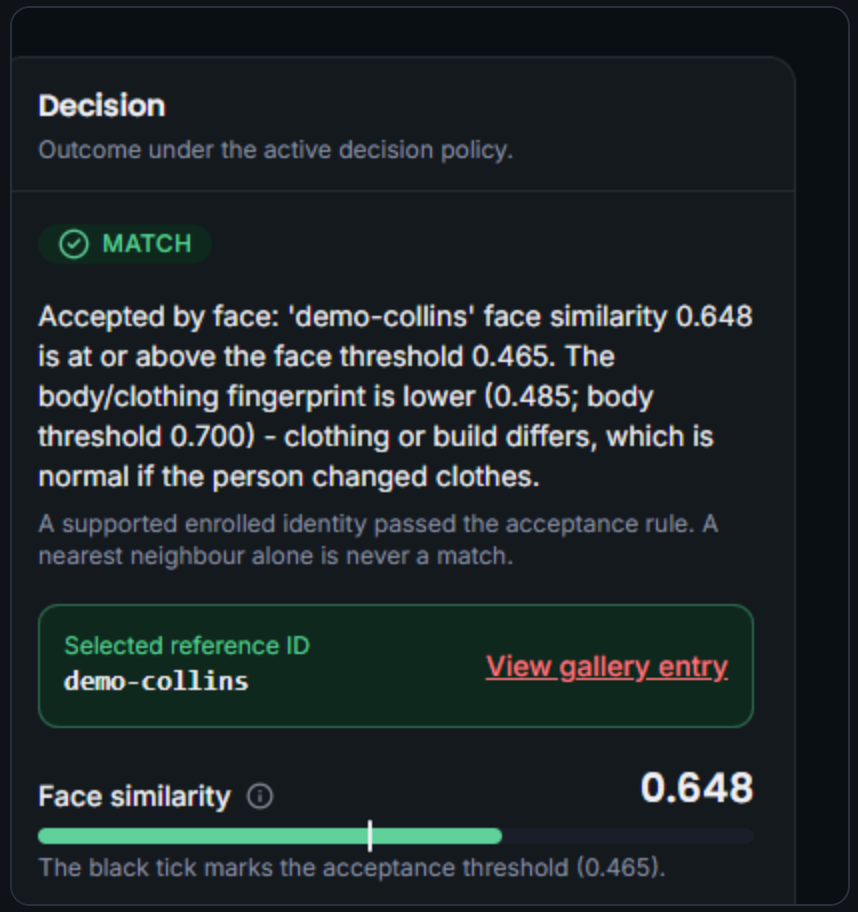
Honest limit: 13 portraits - a demonstration of the mechanism, not a benchmark or an error rate. Run it yourself: **Run the example** on the home page.



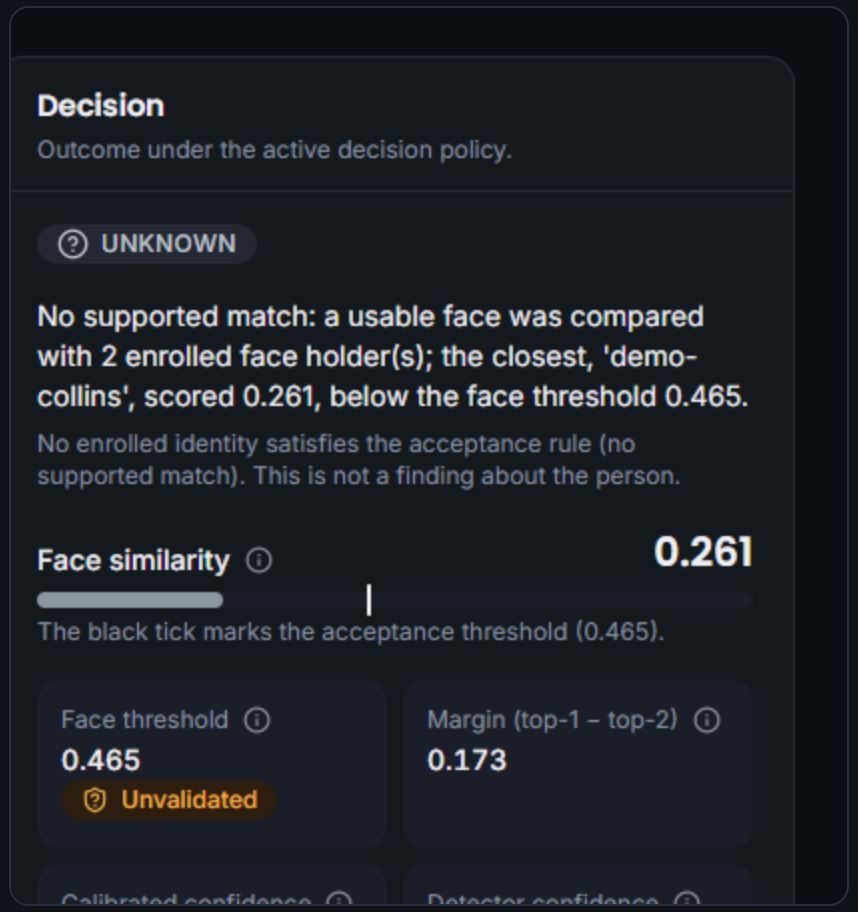
The home page after "Run the example": table, two score plots, and a slider for the face line

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Real photos: the same person, years and clothes apart



MATCH different clothes, 12 years later: **the face decides**



UNKNOWN a stranger: face far below the line

Same 8 public-domain photos

3 of 8
right with clothing only

8 of 8
right with face + body

Public-domain NASA photographs. A small demonstration, not a benchmark.

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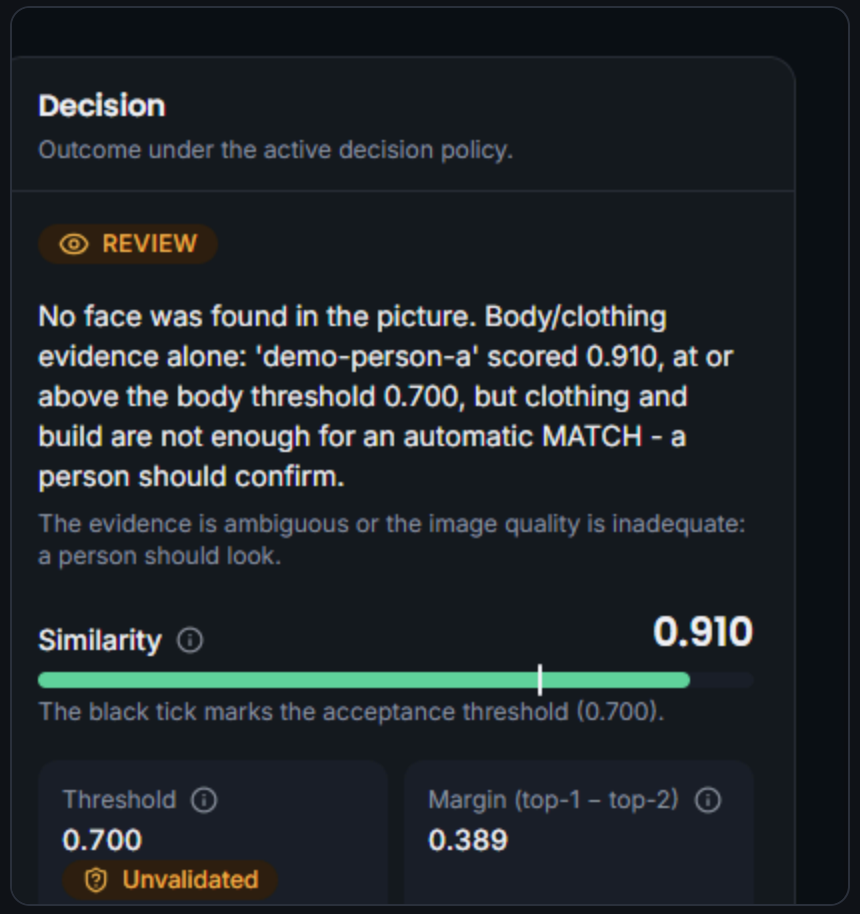
No usable face? It does not pretend.

Clothing matches strongly (0.91) - but the head is hidden, so only clothing and build can be compared.

Clothing alone is weak evidence, so the system does **not** auto-match. It says **REVIEW** and shows why:

> *"No face was found... clothing and build are not enough for an automatic MATCH - a person should confirm."*

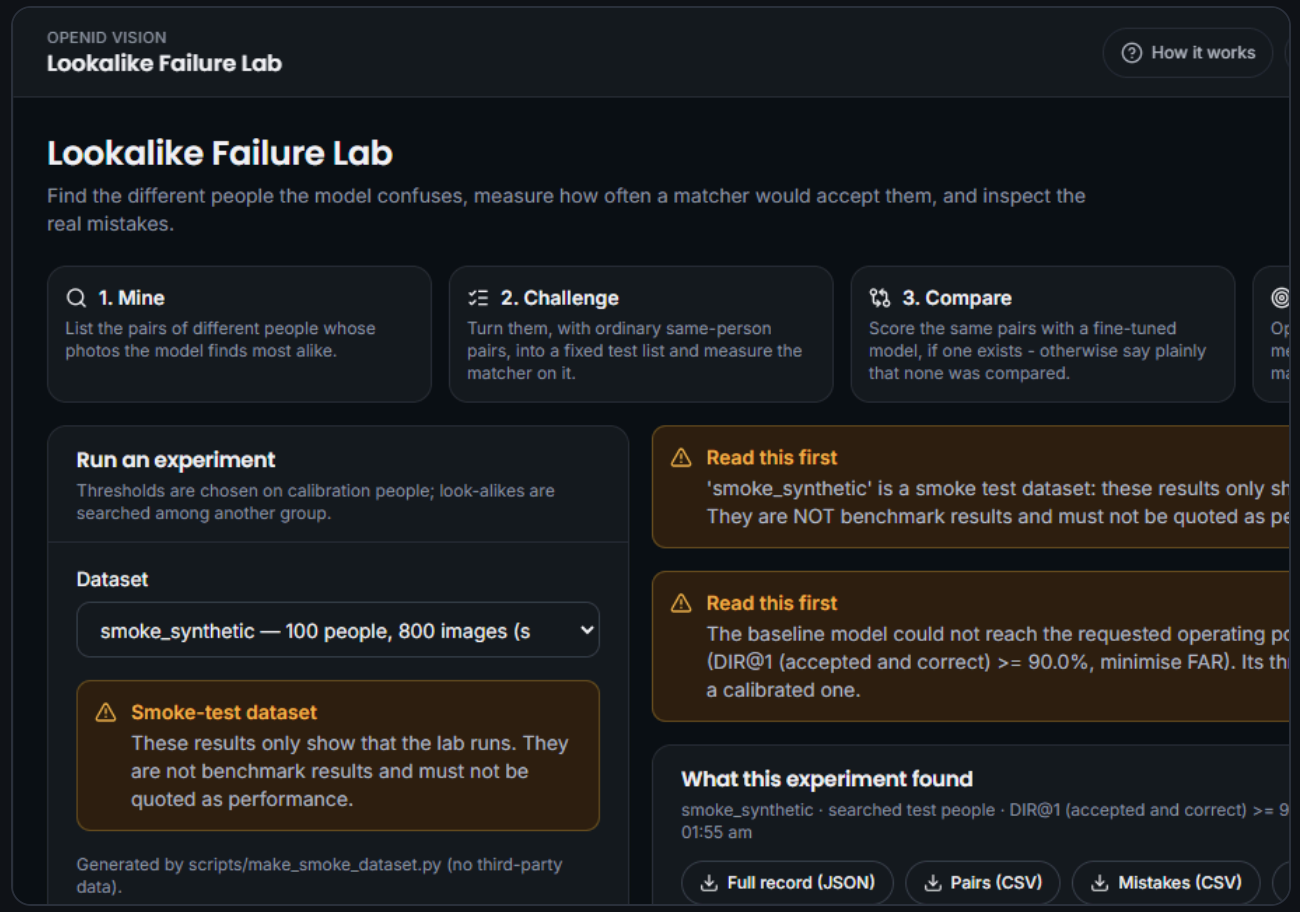
Blurry, tiny, cut-off or turned faces are handled the same way: reported, never silently trusted.



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The Lookalike Failure Lab - finding the mistakes that matter

- Finds the **pairs of different people** the model confuses most
- Builds a **reproducible challenge set** and prints the realistic error rate next to the worst-case one
- Opens each real mistake: **observed numbers** vs **possible causes - labelled hypotheses, never conclusions**
- Compares a baseline with an improved model - or says plainly that none exists
- **Threshold sandbox**: try any threshold safely; it can never change the live system



The screenshot is the synthetic smoke-test set (cartoon figures): it proves the lab works, not how accurate the system is.

What it cannot do - said up front

- **Not a phone's Face ID**: no depth sensor, no liveness check - a photograph can fool a 2-D matcher
- **Twins and look-alikes** can still score high - that is what the review band and the Lab are for
- **Masks, sunglasses, strong angles, low light, tiny faces** make a face unusable - reported, not guessed
- Accuracy measured on a **standard easy benchmark**; other cameras, ages and demographic groups are **not measured**
- The **pedestrian benchmark (Market-1501) ran only on a subset, body only**, with the project's own protocol: modest false-accept rates at high recall; face + body was **not** benchmarked there
- Thresholds are evidence-based defaults, **unvalidated** for any specific camera

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Privacy and consent

- Only **enrolled, consenting** people are ever matched; enrolment needs an explicit confirmation
- A face fingerprint is **biometric data**: stored on this computer only, until the person is deleted
- **Deleting a person erases** their photos, face and body fingerprints; query photos expire after 24 hours
- No cloud service, no scraping, no camera feed; unknown people are never saved, grouped or tracked
- One switch (`REID_FACE_MATCHING=false`) returns to the body-only system; every rule change is versioned and audited
- Biometric data is regulated (GDPR, India's DPDP Act, Illinois BIPA): a real deployment needs a legal basis and an impact assessment

Next steps - and questions

What we would do next

1. Repeat the benchmark with the **official Market-1501 protocol**, on a second dataset, and benchmark **face + body** on footage with usable faces
2. **Calibrate** both thresholds on the real cameras
3. Add **liveness / anti-spoofing**
4. Measure accuracy by **lighting, age and demographic group**
5. Repeat the metric-learning fine-tune on more seeds (one run so far: Rank-1 83.1 → 89.7 %)

Everything is reproducible

431 backend + 157 interface tests · one script per claim:

```
evaluate_face_lfw.py
verify_face_vs_body.py
demo/run_demo.py
demo/run_open_set_example.py
```

If asked something we have not measured: *"I don't know - here is how we would measure it."*